AI 101 for Mechanical Engineers · Module 14

Write an engineering prompt

Lesson 14.1

What you will learn

> Write an engineering prompt with task, audience, system, values, units, assumptions and output; verify a simple photovoltaic calculation; and separate instantaneous power from accumulated energy.

Lesson overview

Module 14 · Lesson 1

Write an engineering prompt

AI 101 for Mechanical Engineers

1 Build the prompt in this order

1. **Task:** the decision or calculation needed
2. **Audience** and **physical system**
3. **Given values**, each with its unit
4. **Assumptions**, e.g. constant power for 2 h
5. **Output:** equations, units, steps, table

Ask it to separate given facts from assumptions and to list missing information.

A good prompt makes checking easier. It does not make the answer true.

2 Check the PV numbers yourself

Authored teaching case: A = 20 m², G = 800 W/m², η = 0.20, η_{inv} = 0.96

$P_{DC} = \eta\, A\, G = 0.20 \times 20\ m^2 \times 800\ W/m^2$ **= 3,200 W**

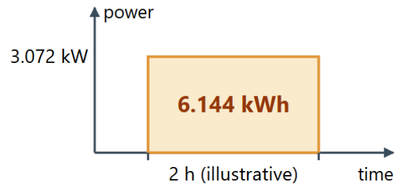
$P_{AC} = \eta_{inv}\, P_{DC} = 0.96 \times 3{,}200\ W$ **= 3,072 W**

Units: m² × W/m² = W; efficiencies have no unit ✓

Spot the error: an answer of "3,200 kW" is wrong by a factor of 1,000, however fluent it sounds.

The values do not describe real equipment performance.

3 Power is not energy



$E = \int P\, dt;\ 3.072\ kW \times 2\ h = 6.144\ kWh$

This is energy over two hours at constant power, **not** a daily yield. A day needs a time profile and losses.

4 Zero-shot and few-shot prompts

Zero-shot
Instructions, context and the required format, with no demonstration examples.

Few-shot
The same, plus one or more demonstration examples, each with an input and its desired output.

Examples clarify the structure you want. They do not verify equations or supply missing physical evidence.

Key takeaway

Specify the task, system, values, units, assumptions and output, then check the answer.
Power is a rate; energy needs a duration or time profile.

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Build an engineering prompt in order

1. Start with the decision or calculation required
2. Name the intended audience and the physical system
3. List every supplied value with its unit
4. State assumptions, e.g. ignoring temperature effects
5. Define the output: equations, units, steps, uncertainty or table

Structure makes checking easier, not the answer true

- Ask the assistant to separate supplied facts from assumptions
- Ask it to identify missing information
- A well-structured prompt does not make the response true

The illustrative array gives 3.2 kW DC

$$P_{DC} = \eta\, A\, G$$

$$P_{DC} = 0.20 \times 20\ \mathrm{m^2} \times 800\ \mathrm{W/m^2} = 3{,}200\ \mathrm{W}$$

- Simplified direct-current power [77]
- A = 20 m², G = 800 W/m², η = 0.20
- An authored teaching case, not real equipment performance

The inverter estimate gives 3.072 kW AC

$$P_{AC} = \eta_{inv} P_{DC}$$

$$P_{AC} = 0.96 \times 3{,}200\ \text{W} = 3{,}072\ \text{W}$$

- Simplified alternating-current estimate
- $m^2 \times W/m^2$ gives W; efficiencies are dimensionless

A claim of 3,200 kW is a unit error

- 3,200 kW contains a factor-of-1,000 unit error
- Fluency and correct engineering units are separate properties
- Calculate independently of the assistant

Power is a rate; energy accumulates over time

Power

- Instantaneous rate
- Measured in watts

Energy

- Time integral of power
- Commonly reported in kilowatt-hours

> One irradiance or power observation cannot determine daily energy.

Two hours at constant power give 6.144 kWh

$$3.072\ \mathrm{kW} \times 2\ \mathrm{h} = 6.144\ \mathrm{kWh}$$

- Explicitly illustrative constant-power assumption, for two hours only
- Must not be presented as a daily yield
- Valid estimate needs time profile, duration, variation and losses

Zero-shot and few-shot prompts differ in examples

Zero-shot

- Task without demonstration examples [67]

Few-shot

- One or more demonstration examples [67]
- Each shows an input and desired output

Both can state context and format. Examples clarify structure; they neither verify equations nor supply evidence.

CHECK YOUR UNDERSTANDING

What is missing when one instantaneous power value is converted into daily energy?

Answer before moving on.

CHECK YOUR UNDERSTANDING

What is missing when one instantaneous power value is converted into daily energy?

> Time information and relevant losses. Use a time profile, or a stated duration with valid constant-power assumptions. In the PV case, P_{DC} is 3,200 W and P_{AC} is 3,072 W.

Key takeaways

- State task, system, values, units, assumptions and output format
- Check equations and units independently of the assistant
- Power is a rate; energy requires integration over time
- Treat numerical examples as illustrative unless physical evidence supports them

References

[67] T. B. Brown et al., “Language models are few-shot learners,” in Adv. Neural Inf. Process. Syst. (NeurIPS), vol. 33, 2020.

[77] J. A. Duffie, W. A. Beckman, and N. Blair, Solar Engineering of Thermal Processes, Photovoltaics and Wind, 5th ed. Hoboken, NJ, USA: Wiley, 2020. doi: 10.1002/9781119540328

Numbers match the course reference list. The written lesson is the authoritative text.